

Zachary Starnes

Software Engineer

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Education

B.S. Computer Science *Clemson University*

08/24 – 05/28

Research

Machine Learning Researcher *Watt AI, Clemson University*

08/25 – 08/26

- Developed and evaluated machine learning models for automated detection of predatory journals.
- Built and evaluated an automated classification pipeline using neural networks and open-source LLMs.
- Presented findings at the 21st Annual Focus on Creative Inquiry Forum and Clemson AI Research Symposium.

Tech Stack Python, PyTorch, Machine Learning, LLMs, Qwen, Llama, Classification

Projects

Yuzic *Open Source Mobile App*

12/24 – Present

github.com/yuzicapp/yuzic

- Built a cross-platform open-source music player for self-hosted media servers, published on the App Store and Google Play.
- Grew the project to 500+ GitHub stars, thousands of users, and 7 outside contributors.
- Designed a modular architecture supporting multiple self-hosted music server backends, with offline playback and native audio.

Tech Stack React Native, TypeScript, Expo, Redux, Express, Jellyfin API, Subsonic API

Reddibase *Machine Learning Pipeline*

04/26 – Present

github.com/eftpmc/reddibase

- Designed a source-neutral pipeline that scrapes, audits, and transforms discussion threads into training data, decoupled from any single platform.
- Served classification and retrieval models behind a FastAPI search endpoint backed by a vector index.

Tech Stack Python, PyTorch, SentenceTransformers, FAISS, FastAPI

Melodari *Web App*

08/24

www.melodari-v2.vercel.app

- Integrated Spotify and other music platform APIs into a unified music library.
- Built profile and social features for playlist sharing across multiple music platforms.

Tech Stack Next.js, Tailwind, Supabase, Google API, Spotify API

Experience

Freelance Software Developer *Various Clients*

04/22 – Present

- Developed full-stack web applications for local businesses using React, Next.js, Node.js, and WordPress.
- Built responsive websites, custom integrations, and content management solutions for multiple clients.

Tech Stack React, Next.js, Node.js, TypeScript, TailwindCSS, WordPress

Junior Developer *Atlas Studios*

10/22 – 06/23

- Collaborated with a small team, including a Google engineer, on an open-world game built in Lua.
- Designed object-oriented systems to improve runtime performance and team ergonomics.
- Contributed to a game serving more than 10,000 players.

Tech Stack Lua, Object-Oriented Design, Game Systems

Technical Skills

- **Languages:** Python, C++, C, Go, Java, JavaScript, TypeScript, Lua, Dart
- **Frameworks:** React, React Native, Next.js, Node.js, Express, PyTorch, Expo, TailwindCSS
- **Tools:** Docker, Git, Linux, MongoDB, MySQL, Redis, Firebase, Supabase, Vercel